FACULTY OF COMPUTING

SEMESTER 1

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SECR1013-DIGITAL LOGIC

SECTION 02

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Acknowledgement

This project is dedicated to our lecturer, Dr. Zuriahati. We would like to express our gratitude towards her for being patient with us and teaching us Digital logic this semester. We got to learn about the digital circuits and components and real-life usage. Therefore, we highly appreciate Dr. Zuriahati for passing the knowledge to us.

We would also like to thank our fellow group members for working hard during the entire project despite having their own schedules and plans. We definitely have to improve our team working skills as long as our problem-solving and thinking skills.

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Background

In this project, we will enhance a lift in a hotel building that supports 8 floors as the previous lift only supported 4 floors. The user of the lift is able to go to the desired floor by inputting a 3 bit number for the desired floor. The elevator is also introduced with a few features which include an access card switch and door condition switch.. The full circuit is drawn using Deeds which uses combinational and sequential circuits.

I. Components of Combinational Circuit:

- 3 bit comparator
- Basic gates
- Switches
- Seven segment display

II. Components of Sequential Circuit:

- 3 bit synchronous counter
- Clock disabler

III. Enhanced features:

- Access card switch
- Door condition switch

Problem Statement

It is a known fact that many unauthorized parties disguised as hotel visitors use the hotel lifts and clearly, this is a threat to the privacy and safety of the customers of the hotel. To avoid any safety issues, we have introduced a hotel visitor authorization scanner in the lifts. Only authorized hotel visitors who have registered themselves at the registration counter get the hotel visitor access card. When hotel visitors enter the lift, they would enter the level button they would like to go to. However, the lift will only close and move when the hotel visitor access card is scanned at the scanner to authorize their visiting pass. Not only is this a safety measure to prevent unknown parties from entering the hotel, moreover, we could also track the movements of visitors in case anything goes wrong as the details of the visitors are registered at the counter. Other than that, we have also changed the previous 2-bit comparator lift to 3-bit comparator which allows the hotel customers and visitors to go up to level 7.

Suggested Solution

State Diagram

The state diagram is drawn based on present state and next state values that is followed by the value of the direction X where value 0 goes clockwise direction and value 1 goes anticlockwise direction. Therefore, it is a saturated counter as count up at maximum value (111) or repeat at the minimum value (000) if count down.

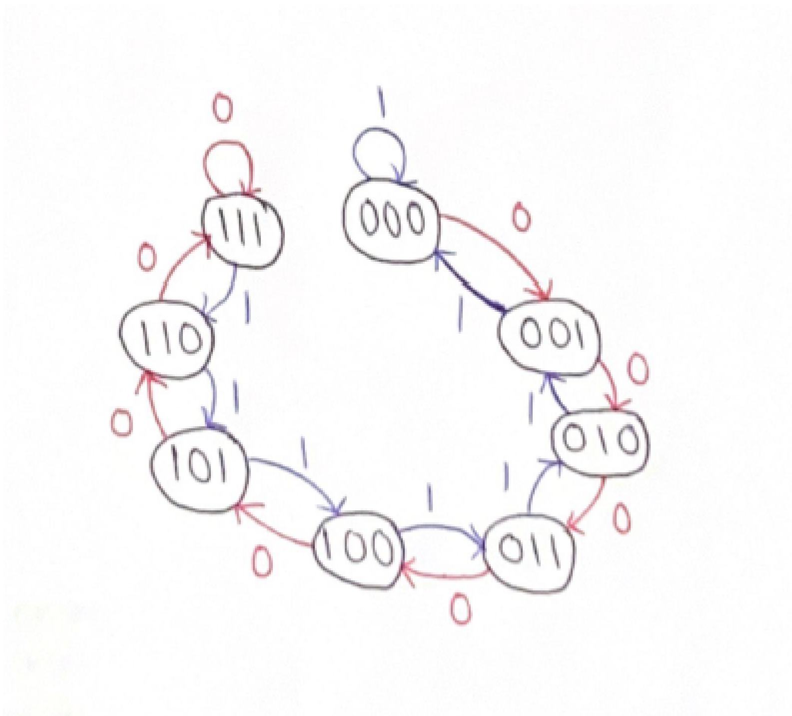


Figure 1

State Table & Transition Table

In this table, next state values are obtained based on the state diagram. The direction X shows count up = 0 and count down = 1. JK flip flop is used based on a 3-bit synchronous counter where FF0 = J0, K0, FF1 = J1, K1 and FF2 = J2, K2 and the values can be obtained by referring to the present state and next state.

Direction, X	Present State			Next State			JK FF					
	Q2n	Q1n	Q0n	Q2n+1	Q1n+1	Q0n+1	J2	K2	J1	K1	J0	K0
0	0	0	0	0	0	1	0	X	0	X	1	X
0	0	0	1	0	1	0	0	X	1	X	X	1
0	0	1	0	0	1	1	0	X	X	0	1	X
0	0	1	1	1	0	0	1	X	X	1	X	1
0	1	0	0	1	0	1	X	0	0	X	1	X
0	1	0	1	1	1	0	X	0	1	X	X	1
0	1	1	0	1	1	1	X	0	X	0	1	X
0	1	1	1	1	1	1	X	0	X	0	X	0
1	0	0	0	0	0	0	0	X	0	X	0	X
1	0	0	1	0	0	0	0	X	0	X	X	1
1	0	1	0	0	0	1	0	X	X	1	1	X
1	0	1	1	0	1	0	0	X	X	0	X	1
1	1	0	0	0	1	1	X	1	1	X	1	X
1	1	0	1	1	0	0	X	0	0	X	X	1
1	1	1	0	1	0	1	X	0	X	1	1	X

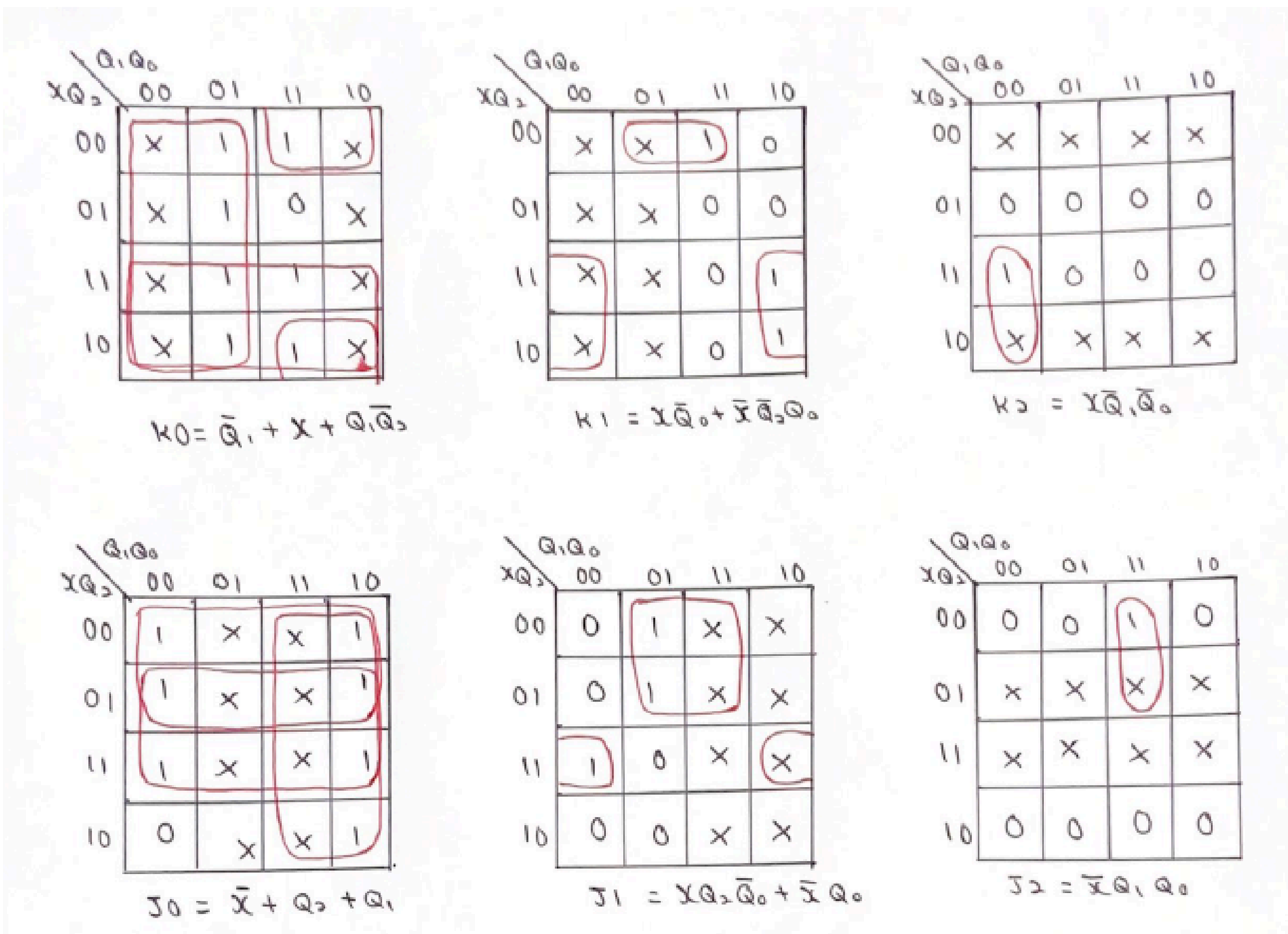
1	1	1	1	1	1	0	X	0	X	0	X	1

Table 1

Karnaugh Map (K-map)

After finishing grouping in K-map, simplified equations for K0, K1, K2, J0, J1, J2 are formed.

Figure 2



System Implementation

From the truth table and Karnaugh Map (K-Map), the equations for all JK flip flops are formed. With these equations, we had built a DEEDS circuit to show how the lift works. First of all, three JK flip flops are placed into the circuit. To obtain the input for each J and K in flip flop, we followed the equation that has been formed from the K-map.

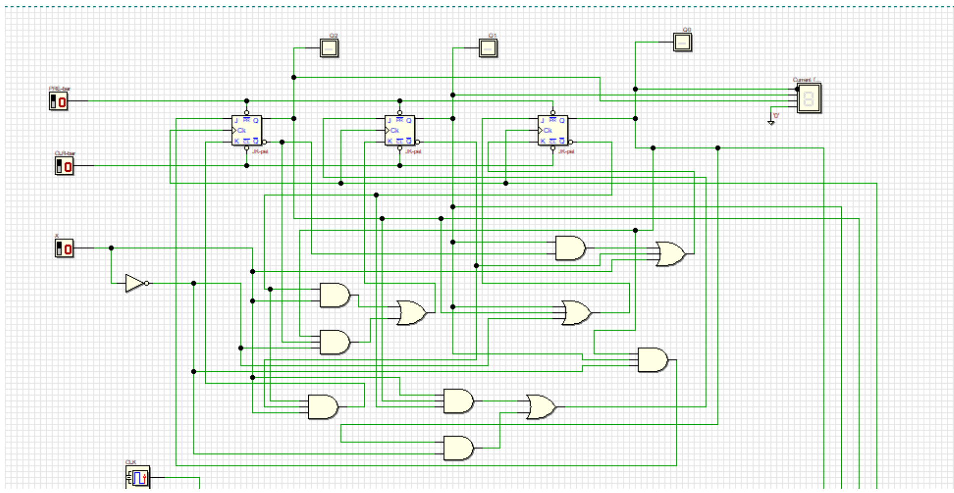


Figure 3

There are few special features to our lift system such as access card switch and door condition switch. These two conditions will decide the functionality of the lift along with power, comparator, clock, door and start switch.

Power:

1 = powered 0 = no power

Comparator:

1 = not the same floor 0 = same floor

Start:

1 = start 0 = stop

Door:

1 = door closed 0 = door opened

Access Card:

1 = access card detected 0 = access card not detected

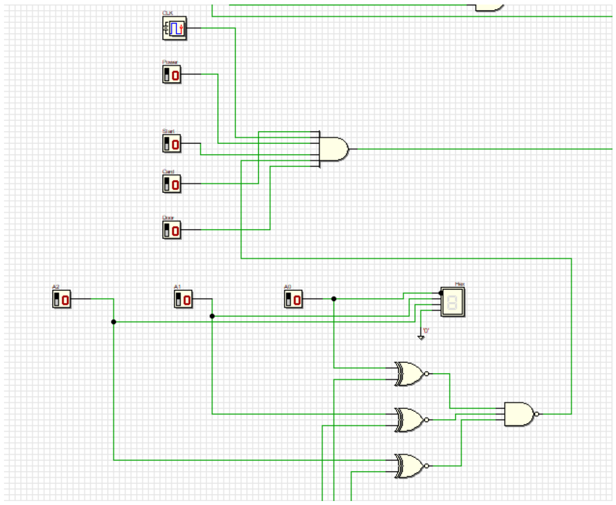


Figure 4

The AND gate decides either the lift will work or not. If all input of the AND gate is equal to 1, the lift will be able to function normally. If otherwise, the lift will not be able to function and users cannot use it.

If the AND gate is now equal to 1, to make the lift be able to run in synchronous mode, the PRESET bar and CLEAR bar must be set into 1.

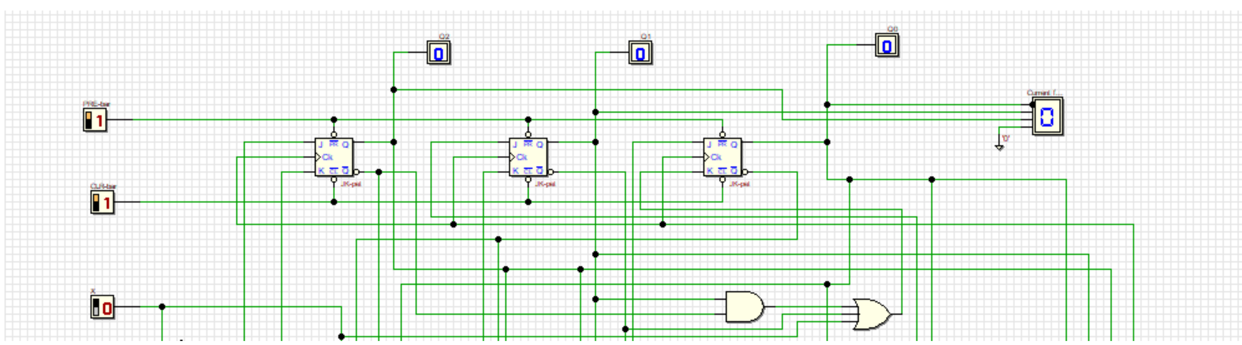


Figure 5

In some scenarios, if the user inputs the same level as the current one, the lift will not move to any other level since it has already arrived at its destination no matter how many times the clock is switched. The same thing also applies when a user arrives at their desired level. The lift will not continue up or down once the current level is the same as the input level. This is because the comparator will detect an equality and send the output 1 which will revert to 0 by the NAND gate.

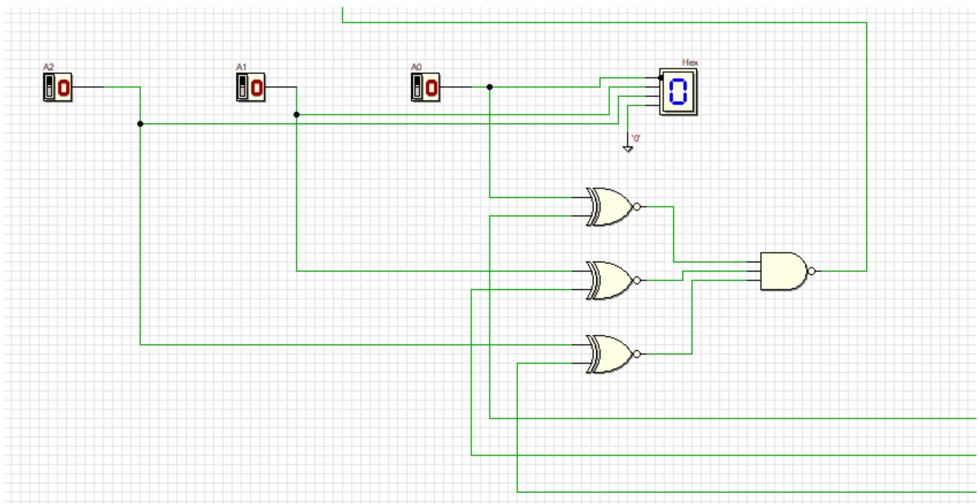


Figure 6

This caused the AND gate that will be activating the CLK for all the flip flops to hold a value 0. The flip flops will be deactivated thus making the lift stop functioning and stop at its current level.

In this lift system, there is an X switch to determine the direction of the lift either up or down.

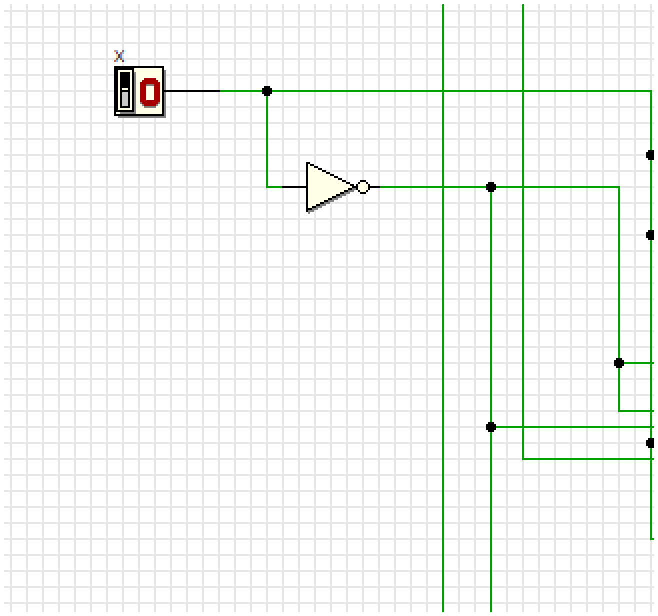


Figure 7

If the direction of the lift is going up, switch X must be 0. The value of X will be 1 if the lift goes the opposite way which is down.

If the user wants to input the level they want to go to, they just have to input the number in binary value in the switch button below near the comparator and the Hex display will show the number key in by the users.

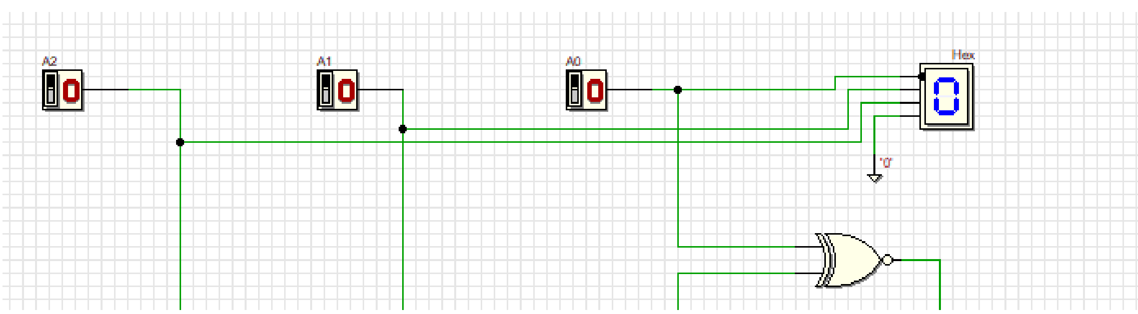


Figure 8

Users can input from level 0 to 7 in binary form. Then, the other Hex display will show which floor the lift is on now so users can know which way to go, either down or up and the X value.

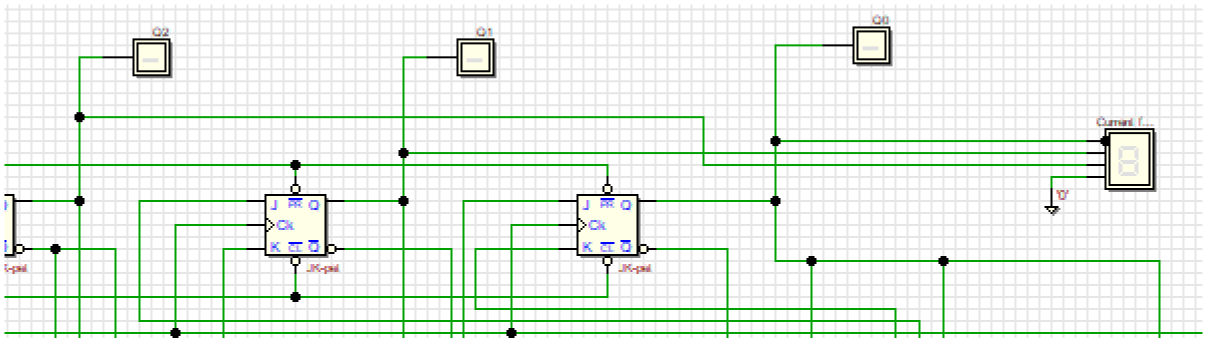


Figure 9

All these components formed a full circuit which let the lift be able to function normally and efficiently thus solving the problem. Below is our lift system full circuit:

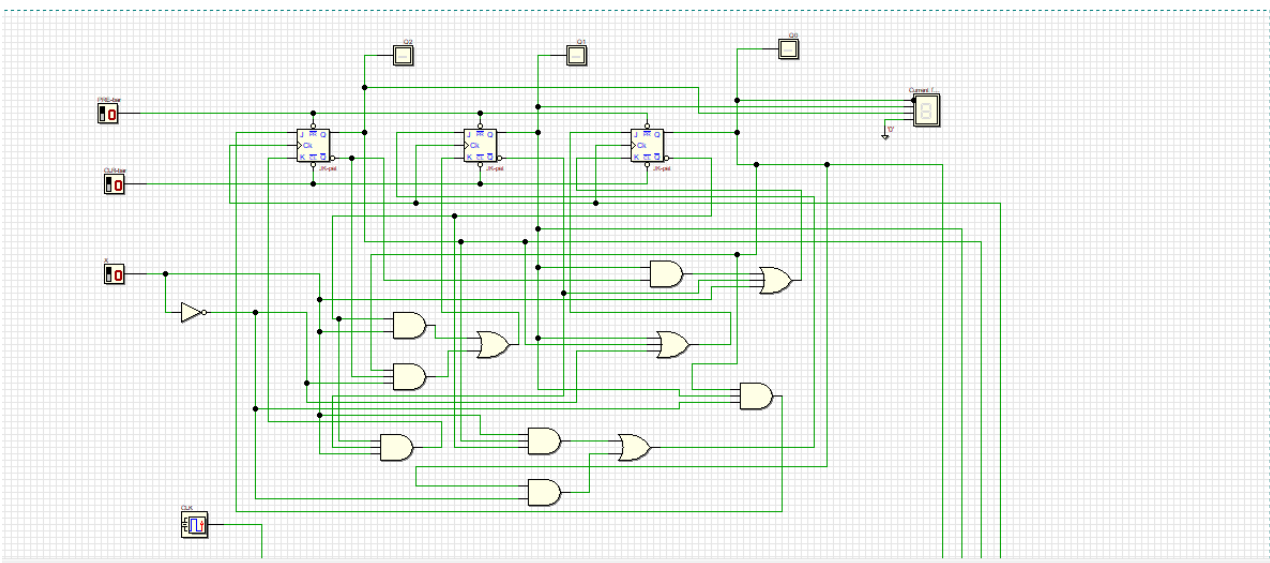


Figure 10

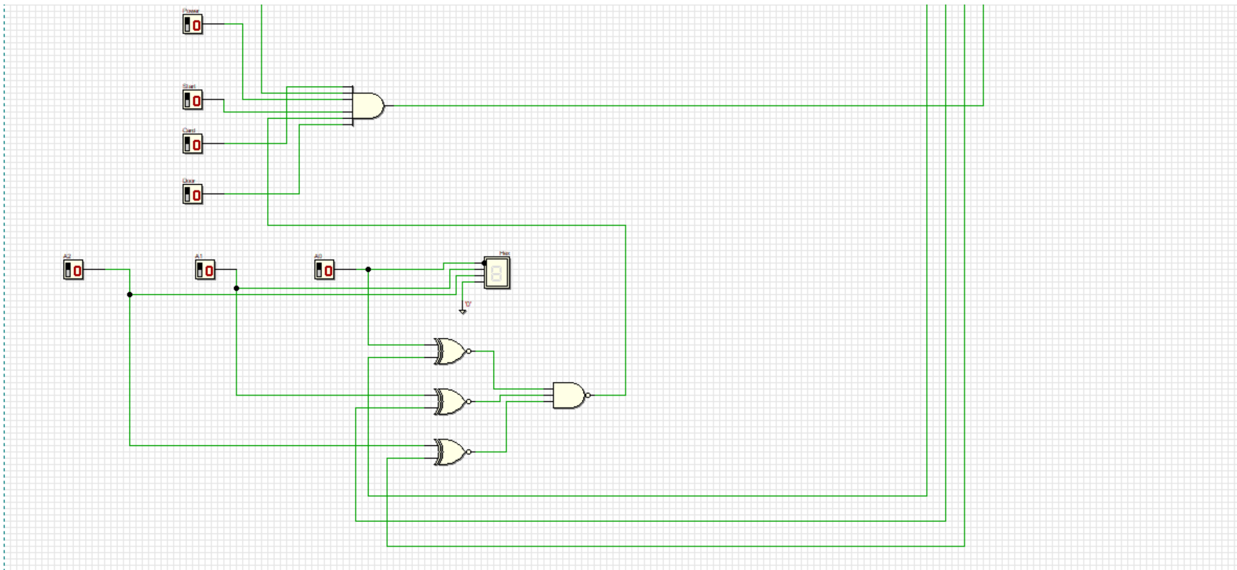


Figure 11

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Conclusion and Reflection

As a conclusion, the solution we have provided is user-friendly and can be easily implemented at all the hotels. In this circuit, the main components are a 3-bit comparator, input switches, the seven segment displays and a 3-bit synchronous counter. Introducing the 3-bit comparator allows the users to go from ground level to level 7 whereas previously they could only go till level 3. The feature that we have mainly implemented is the access card switch and the door condition switch. The access card switch and the door condition switch are interrelated where the door condition switch works only if the access card switch is HIGH.

Gladly, we didn't face any issues while we worked on this project. We are grateful for being assigned this task as we get to implement our knowledge on real-life problems and find a solution for it. We are also looking forward to implementing our knowledge gained from the Digital Logic course in our upcoming courses in the following semesters.

References

Abd Brahim et al. (2023). *Digital Logic Fifth Edition 2023*. Faculty of Computing, Universiti Teknologi Malaysia

Appendices

Group Members:

1) Nurul Ika Syafiny Binti Azhar
2) Nuraisyah Binti Mohd Zikre
3) Harini A/P Sangaran
4) Saranya A/P Jayarama Reddy

Responsibilities:

1) **Nurul Ika Syafiny Binti Azhar**
    - System Implementation
    - References

2) **Nuraisyah Binti Mohd Zikre**
    - Front Cover
    - Background/Overview

3) **Harini A/P Sangaran**
    - Acknowledgement
    - Problem Statement
    - Conclusion and Reflection

4) **Saranya A/P Jayarama Reddy**
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    - Suggested Solution
    - Appendices

Group Discussion in Whatsapp Group

Project discussion was done in the whatsapp group and we have divided our work to complete the project.

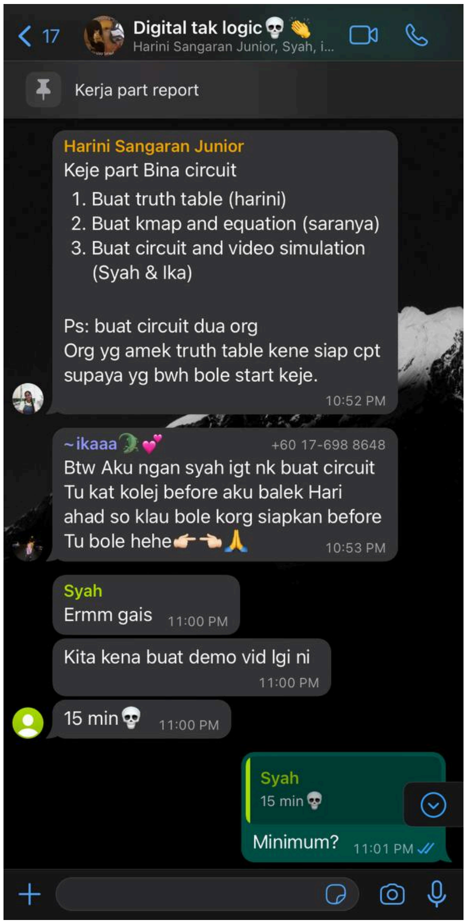

Figure 12

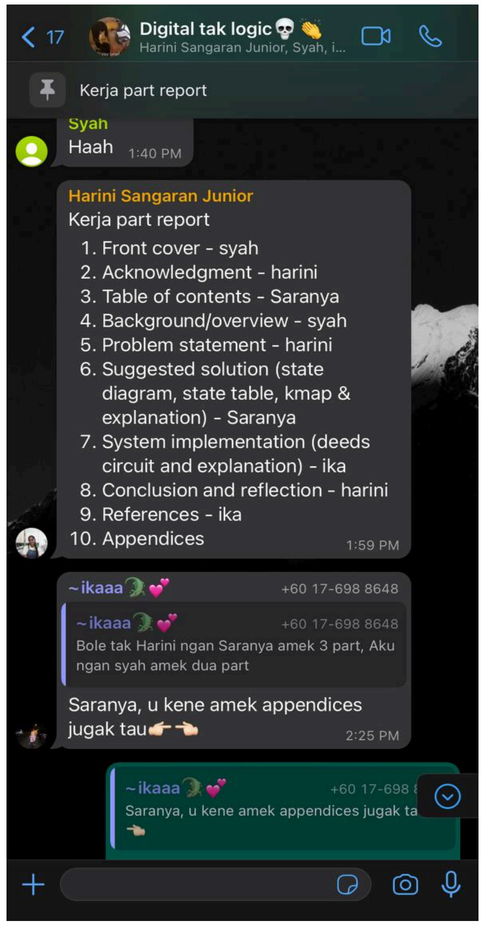

Figure 13

Figure 14